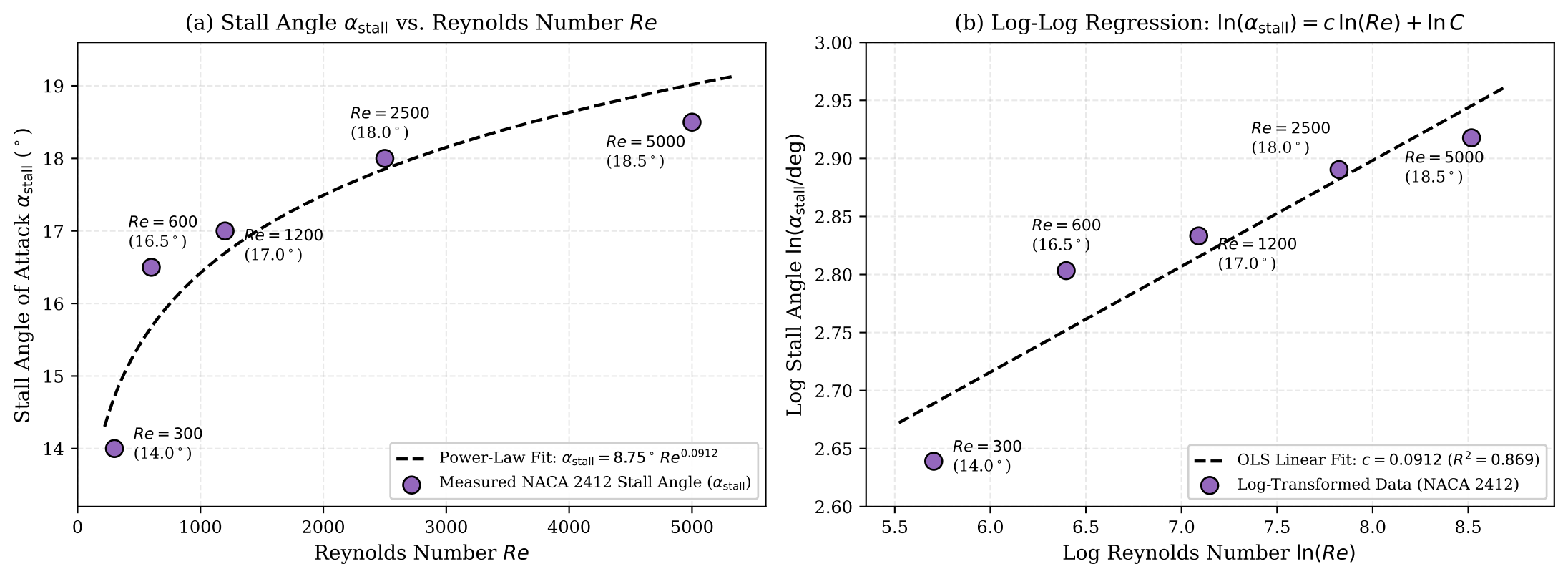
(a) Stall Angle α_{stall} vs. Reynolds Number Re
Stall Angle of Attack α_{stall} (°)
Reynolds Number Re
$Re = 300$ (14.0°)
$Re = 600$ (16.5°)
$Re = 1200$ (17.0°)
$Re = 2500$ (18.0°)
$Re = 5000$ (18.5°)
Power-Law Fit: $\alpha_{stall} = 8.75° \, Re^{0.0912}$
Measured NACA 2412 Stall Angle (α_{stall})
(b) Log-Log Regression: $\ln(\alpha_{stall}) = c \ln(Re) + \ln C$
Log Stall Angle $\ln(\alpha_{stall}/\text{deg})$
Log Reynolds Number $\ln(Re)$
$Re = 300$ (14.0°)
$Re = 600$ (16.5°)
$Re = 1200$ (17.0°)
$Re = 2500$ (18.0°)
$Re = 5000$ (18.5°)
OLS Linear Fit: $c = 0.0912$ ($R^2 = 0.869$)
Log-Transformed Data (NACA 2412)